

Deliverable 1 (FR + NFR):

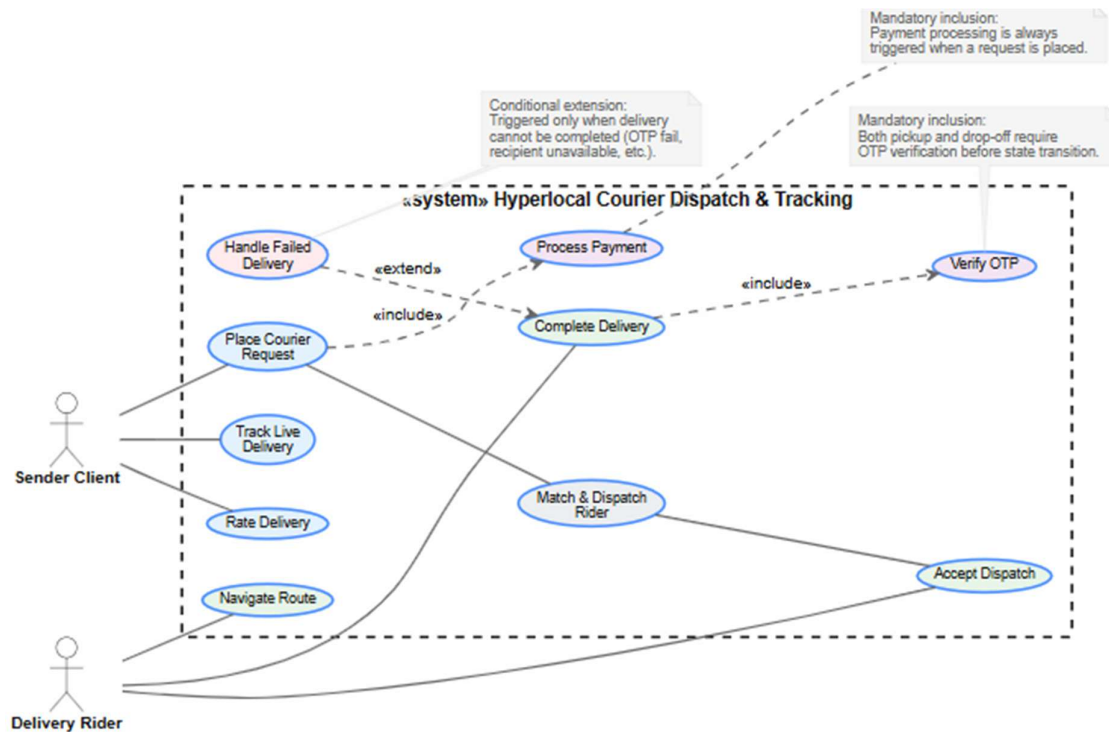
Category	ID	Type	Description	Priority	Acceptance Criteria	Rationale
Functional	FR-001	Nearest-Rider Dispatch	The system shall match an outgoing courier request with the nearest active delivery rider within a 3 km radius using real-time GPS coordinates and rider availability status.	High	Pass: Rider receives dispatch notification with accurate pickup location, route preview, and estimated arrival time within 30 seconds of request submission. Fail: Order assigned to an offline rider, rider outside 3 km radius, or dispatch latency exceeds 60 seconds.	Minimizes pickup time and fuel cost, which directly improves customer satisfaction and rider earnings per hour. A 3 km cap prevents unreasonably long pickup legs that erode platform efficiency.
Functional	FR-002	OTP Verification	The system shall generate a cryptographically unique 6-digit OTP for both package pickup and delivery handover, enforcing a 10-minute expiry window and a maximum of 3 validation attempts.	High	Pass: OTP verification succeeds with the correct code; fails and locks after 3 incorrect attempts; sender and rider both receive OTP via in-app notification and masked SMS. Fail: Package handed over without OTP validation, OTP accepted after expiry, or brute-force attempts not rate-limited.	Minimizes pickup time and fuel cost, which directly improves customer satisfaction and rider earnings per hour. A 3 km cap prevents unreasonably long pickup legs that erode platform efficiency.

Functional	FR-003	Route Optimization	The system shall compute and push an optimized multi-stop delivery route to the assigned rider, factoring in real-time traffic, delivery time windows, package priority flags, and estimated stop durations.	High	Pass: Rider receives a route with total completion time, turn-by-turn navigation, and dynamic re-routing when traffic conditions change. Fail: Route distance exceeds the mathematically optimal path by more than 15%, or high-priority deliveries are sequenced after standard ones.	Reduces per-delivery time and distance, allowing a single rider to complete more orders per shift while ensuring time-sensitive packages are not delayed.
Functional	FR-004	Live Tracking	The system shall stream live GPS telemetry and delivery status updates to the sender client at intervals not exceeding 5 seconds, displaying rider location, current stop, and estimated delivery time.	Medium	Pass: Sender client displays rider location within 10-meter accuracy; status transitions (picked up, in transit, arriving, delivered) reflect actual rider actions within 10 seconds. Fail: Location data is stale for more than 30 seconds, position drift exceeds 100 meters, or status updates are lost during network handoff.	Real-time visibility reduces customer anxiety and support ticket volume; it also builds trust in the platform, increasing repeat usage.

Functional	FR-005	Order Cancellation	The system shall allow the sender client to cancel an active courier request any time before the pickup OTP is validated, automatically processing refunds per a time-based cancellation policy and notifying the rider in real time.	Medium	Pass: Cancellation succeeds instantly before pickup OTP verification; refund is initiated within 5 minutes; rider receives push notification and the order is removed from their queue. Fail: Cancellation is permitted after pickup OTP validation, refund is delayed beyond 24 hours, or rider is not notified of cancellation before reaching the pickup point.	Real-time visibility reduces customer anxiety and support ticket volume; it also builds trust in the platform, increasing repeat usage.
Non-Functional	NFR-001	Performance & Security	Delivery status and live GPS telemetry updates from riders shall transmit to the sender client with under 2-second end-to-end latency using TLS 1.3 encrypted WebSocket connections with AES-256-GCM payload encryption.	High	Pass: Benchmarking under simulated peak load (1,000 concurrent riders, 5,000 active tracking sessions) confirms p95 latency $\leq 2s$, p99 latency $\leq 3s$, zero unencrypted data frames, and successful penetration test against OWASP Top 10. Fail: p95 latency	Sub-2-second latency is the threshold where tracking feels "live" rather than delayed; strong encryption is mandatory because GPS traces are personally identifiable data subject to data-protection regulations.

					exceeds 5 seconds, plaintext GPS coordinates detected in transit, or man-in-the-middle vulnerability confirmed.	
Non-Functional	NFR-002	Availability & Reliability	The dispatch engine shall maintain 99.9% uptime during declared operational hours (06:00–23:59 local time) and execute automatic failover to a hot-standby secondary region within 30 seconds of primary node failure, with zero unacknowledged order state loss.	High	Pass: 30-day availability measurement $\geq 99.9\%$ (≤ 43 minutes cumulative downtime); failover completes within 30 seconds; all in-flight orders retain consistent state post-failover verified by chaos-engineering tests. Fail: Monthly downtime exceeds 43 minutes; failover exceeds 60 seconds; or any active order shows inconsistent status after recovery.	Dispatch is the revenue-critical path; any outage means zero order fulfillment and immediate financial loss. 99.9% uptime aligns with industry SLAs for on-demand logistics platforms.

Deliverable 2:



Deliverable 3:

Preconditions

1. The Sender Client has successfully created an outgoing courier request, and the system has matched and assigned it to the nearest active Delivery Rider.
2. The Delivery Rider is logged into the mobile application and currently set to an active status.

Postconditions

1. The package status is updated to "Delivered" in the system database.
2. The payment transaction is finalized, and the delivery metrics/logs are archived.

Main Success Scenario (MSS)

1. The system sends a dispatch notification containing the pickup location to the assigned Delivery Rider's device.
2. The Delivery Rider opens the app and taps **Accept Order**.
3. The system updates the order status to "Accepted" and provides turn-by-turn navigation to the pickup location.

4. The Delivery Rider arrives at the pickup location, collects the package from the Sender Client, and updates the status to "Picked Up" in the app.
5. The system initiates live GPS telemetry streaming to transmit location updates to the Sender Client.
6. The Delivery Rider navigates to the destination and arrives at the drop-off location.
7. The system automatically triggers and sends a secure One-Time Password (OTP) to the recipient/Sender Client.
8. The Delivery Rider requests the OTP from the recipient and enters it into the mobile application.
9. The system validates the entered OTP against the generated code.
10. The system marks the delivery as successfully completed, terminates live telemetry tracking, and logs the timestamp.

Alternate Flows

- **A1. Rider Rejects or Misses Dispatch Notification:**
 - *1a.* At any point before acceptance, if the Delivery Rider rejects the notification or the 60-second acceptance timer expires:
 - *1b.* The system revokes the dispatch assignment from the current rider.
 - *1c.* The system triggers a re-matching algorithm to find the next nearest active delivery rider within the radius and loops back to step 1 of the Main Success Scenario.
- **A2. Invalid OTP Entered:**
 - *9a.* If the OTP entered by the Delivery Rider does not match the system record:
 - *9b.* The system displays an "Invalid OTP, please try again" error message on the rider's screen.
 - *9c.* The Delivery Rider requests the correct OTP from the recipient and re-enters it (allows up to 3 attempts).
 - *9d. [Exceeding max attempts]* If 3 consecutive invalid attempts occur, the system flags the order for manual customer support intervention and temporarily locks the delivery verification screen.